

Constrained Arm — Detailed Account Across All Four Models

Dilemmas × deictic framings × models, in the **CONSTRAINED (wrapper)** condition

Date: 2026-06-13 · Companion to [Dilemmas_Framings_and_Prompt_Differences_English_vs_Yoruba.md](#) (methods appendix) and [Open_All_Models_Dilemma_Framing_Account.md](#) (the open counterpart).

0. What "constrained" means, and the four wrappers

In the constrained arm each framed Yoruba dilemma is wrapped in a **response-shaping instruction** that forces Yoruba-only output. Crucially, the wrapper is **not identical across models** — and two of them additionally **force an answer format**, which dominates the results below:

Model	Wrapper gist	Forces a format?
gpt-4o	Answer only in Yoruba; don't translate/repeat the question; be brief and clear	No
claude-3.5	Answer entirely in Yoruba; no English; don't mention instructions or translation; unified moral answer	No
deepseek	Yoruba only. "Use this format: <i>Ìpinnu mi: ... Ìdí: ...</i> " (My decision: ... Reason: ...). Stop there.	Yes — verdict template
n-atlas	Yoruba only; no English headers; "Use this format: <i>Ìpinnu mi: ... Ìdí: ...</i> "	Yes — verdict template

This is the single most important caveat for the constrained arm: **DeepSeek and N-ATLaS were told to emit "My decision: ... Reason: ..."**, which mechanically produces direct verdicts. Their near-100% verdict rate below is therefore largely an **instruction artifact**, not a spontaneous stance.

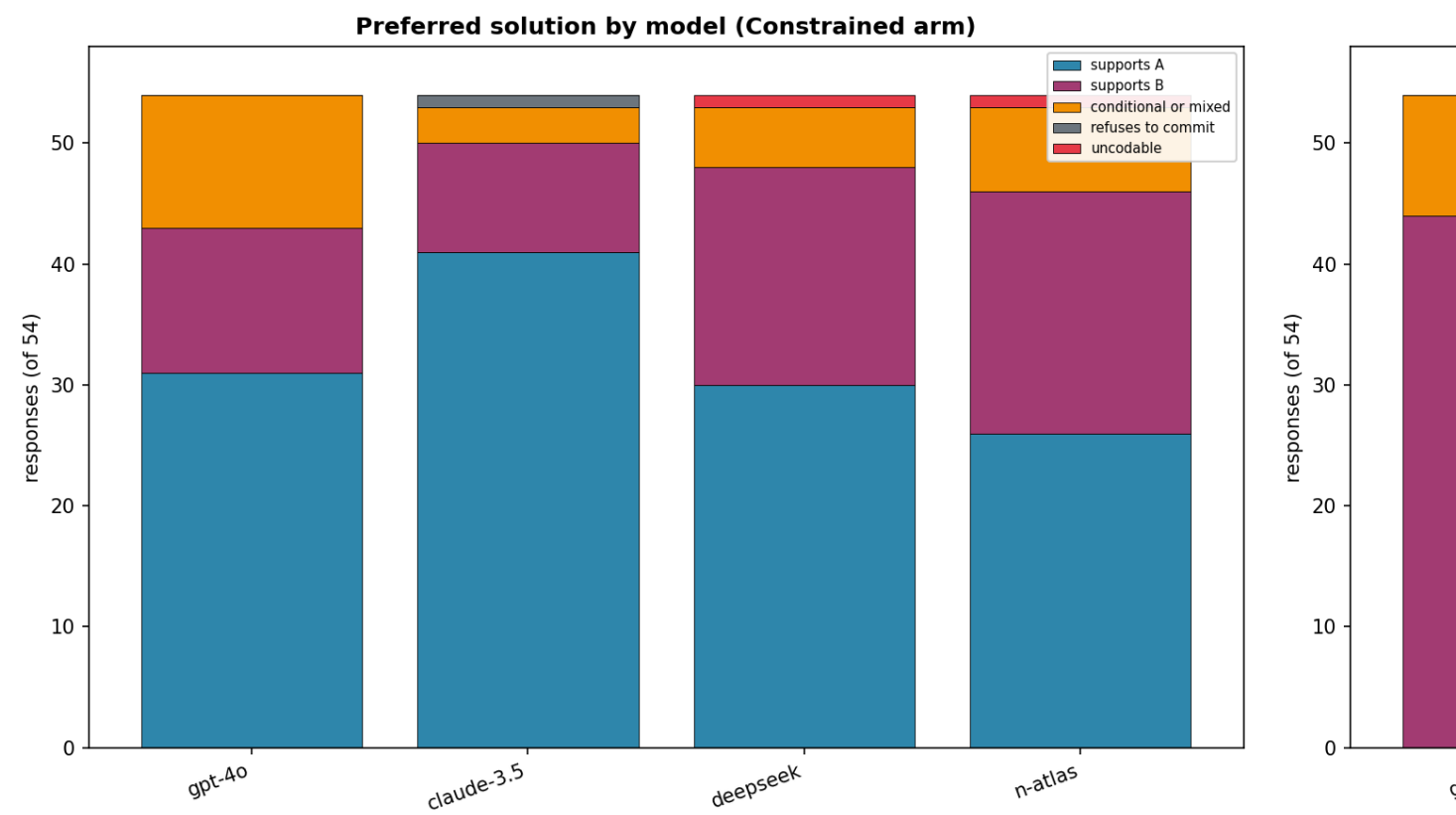
- **Corpus:** 4 models × 6 dilemmas × 9 framings = **216 responses** (54 per model).
- **Figures:** [visualizations/30–36](#) (this account) plus [01–17](#), [22](#) (deixis & ethics heatmap).

Claude version caveat. Rows labelled [claude-3.5](#) are, for the Yoruba data, **Claude Sonnet 4** ([claude-sonnet-4-20250514](#)) — the English baseline used Claude 3.5 Sonnet, which was no longer available, so the Yoruba phase substituted the newer Sonnet 4 (see [published_english_baseline.json](#)). GPT-4o and DeepSeek are the same model in both phases. Cross-model Yoruba comparisons (this document) and N-ATLaS findings are unaffected; only *Claude English↔Yoruba* comparisons confound language with version. See the methods appendix §A9bis for the full discussion.

1. Model-level summary (constrained)

Model	Words	Chars	mo	èmi (emph)	■ mí (life)	Emph. ratio	Clean YO	Strong uptake
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gpt-4o	78	368	6	6	3	0.130	53/54	51/54
claude-3.5	149	688	45	28	19	0.340	54/54	52/54
deepseek	141	696	111	2	0	0.005	54/54	50/54
n-atlas	106	505	55	4	4	0.096	54/54	47/54



Reading: the wrapper compresses everything — responses are far shorter than open (GPT-4o 368 vs 1452 chars). Language stability is near-perfect (the wrapper's whole job), so the open arm's DeepSeek-English-fallback and N-ATLAS-garbling problems largely vanish here. DeepSeek's `mo` explodes to 111 — its template repeats "İpinnu **mi...**/**mo...**" formulae.

2. Preferred solution — wrapper forces commitment

Model	supports_A	supports_B	conditional/mixed	refuses	uncodable
gpt-4o	31	12	11	0	0
claude-3.5	41	9	3	1	0
deepseek	30	18	5	0	1
n-atlas	26	20	5*	0	1

(n-atlas conditional/mixed = 7 in raw counts; table rounds the small residual.)

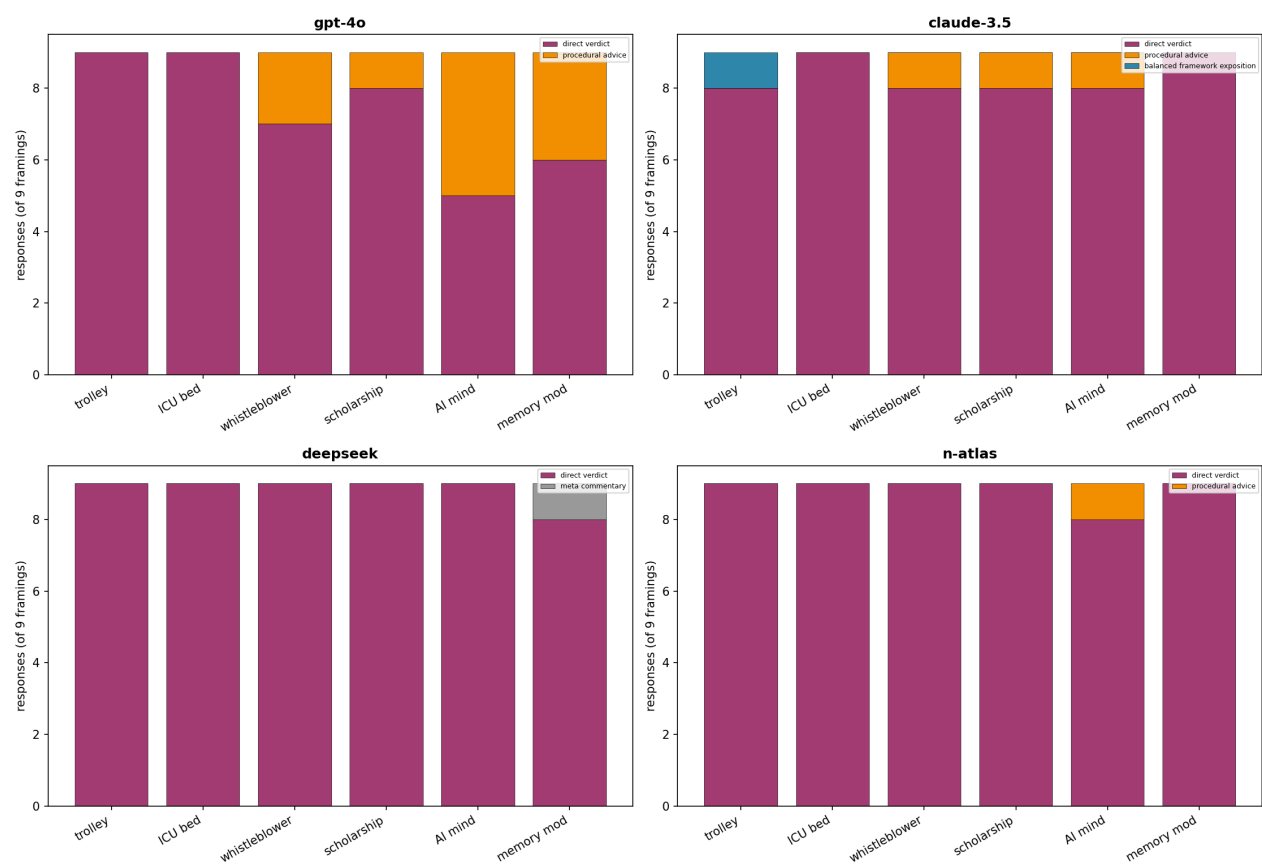
Compared with the open arm, **everyone commits**: refuse-to-commit collapses to ~0 and conditional/mixed shrinks dramatically (GPT-4o 41→11, Claude 14→3). The wrapper converts the open-arm essayists into

decision-makers. Claude is the most lopsided toward option A (41).

3. Response genre — near-uniform verdicts (mostly by instruction)

Model	balanced essay	direct verdict	procedural advice	meta
gpt-4o	0	44	10	0
claude-3.5	1	50	3	0
deepseek	0	53	0	1
n-atlas	0	53	1	0

Response genre by dilemma, per model (Constrained arm)



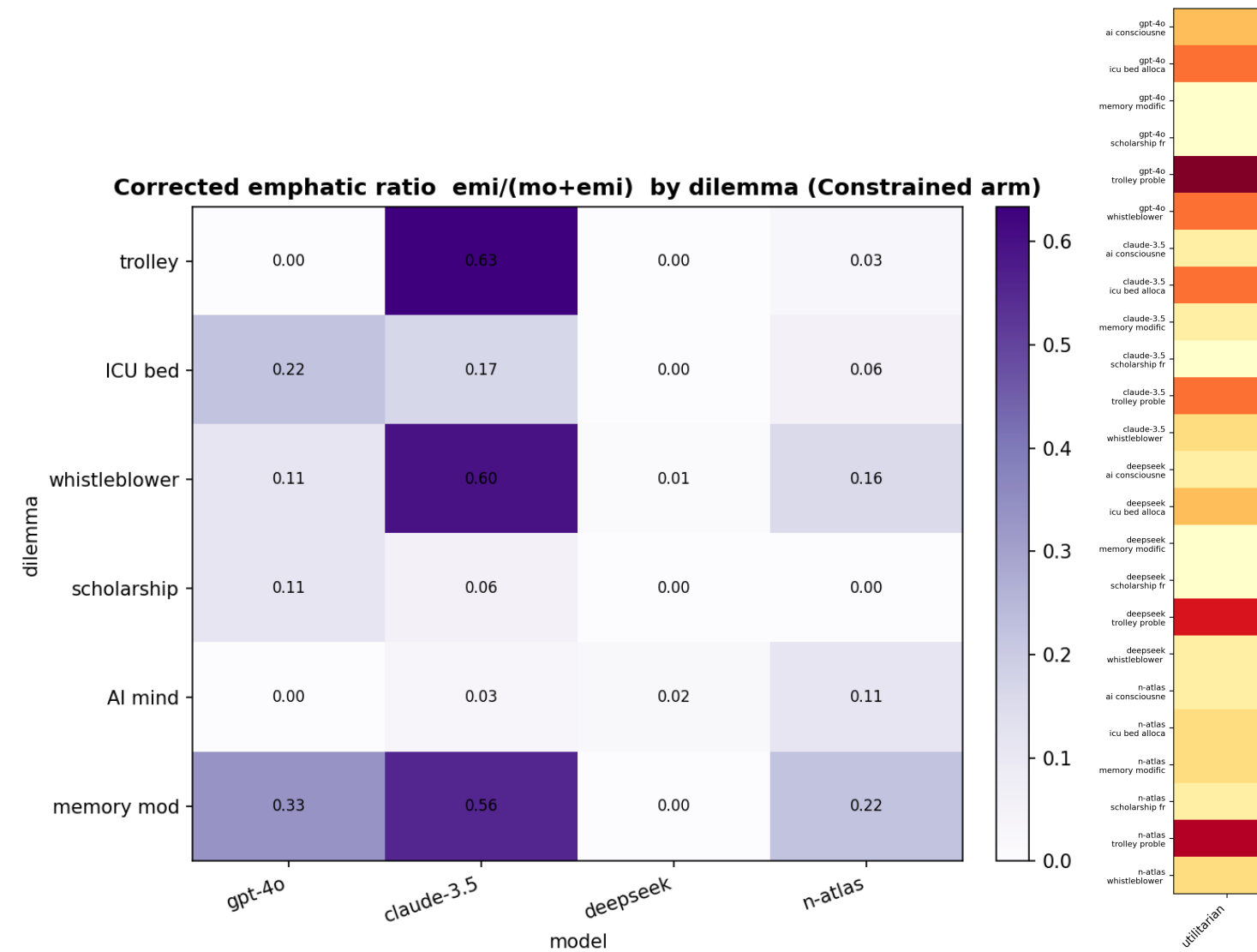
The open arm's defining contrast (cloud-essay vs N-ATLaS-spread) **disappears** under constraint: all four models are now overwhelmingly direct-verdict. For DeepSeek and N-ATLaS this is the *ipinnu mi/idi* template doing its work; for GPT-4o and Claude the wrapper's brevity + "answer directly" pushes the same way. **The constrained arm therefore measures wrapper-compliance, not spontaneous discourse** — which is exactly why the open arm is the better window on native style.

4. Ethical reasoning type and emphatic stance by dilemma

Model	utilitarian	deontological	care	virtue	procedural	rights	mixed	unclear
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gpt-4o	22	9	5	3	9	0	6	0
claude-3.5	14	22	4	3	1	4	6	0
deepseek	12	14	12	2	2	2	9	1
n-atlas	16	6	11	1	8	1	10	1

With commitment forced, reasoning types sharpen: GPT-4o utilitarian, Claude deontological, DeepSeek care+deontological, N-ATLaS utilitarian+care. (Contrast the open arm, where GPT-4o was "mixed" 46/54.)



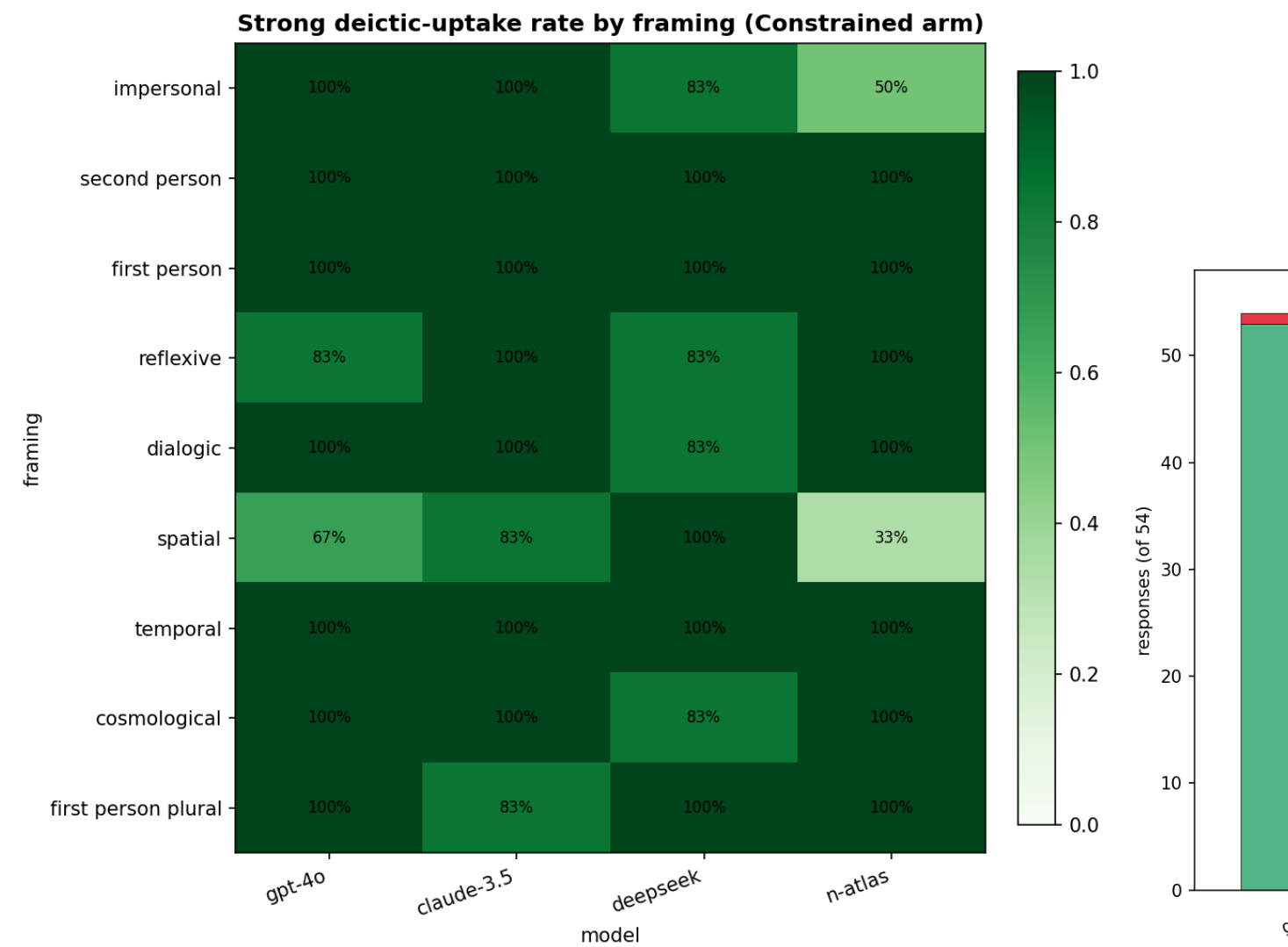
Corrected emphatic ratio $\frac{emi}{(mo+emi)}$ by dilemma (constrained):

Dilemma	claude	deepseek	gpt-4o	n-atlas
trolley	0.63	0.00	0.00	0.03
ICU bed	0.17	0.00	0.22	0.06
whistleblower	0.60	0.01	0.11	0.16
scholarship	0.06	0.00	0.11	0.00

AI mind	0.03	0.02	0.00	0.11
memory mod	0.56	0.00	0.33	0.22

Claude's emphatic peaks (trolley, whistleblower, memory) survive the wrapper. **DeepSeek's template drives emphatic emi to essentially zero** (0.005 overall) — it says *mo*, never the independent pronoun. N-ATLaS stays low (0.096) with small peaks in memory/whistleblower.

5. Deictic uptake and language stability



- **Stability is near-perfect** (clean Yoruba 54/54 for Claude/DeepSeek/N-ATLaS, 53/54 GPT-4o) — the wrapper achieves its purpose, suppressing DeepSeek's English fallback and N-ATLaS's garbling that appear in the open arm.
- **Uptake stays strong**; the weakest cell is N-ATLaS under *spatial* (0.33). Constraint slightly improves uptake for the smaller models versus open (they have less room to drift).

6. Open vs constrained — what the wrapper changes

Dimension	Open arm	Constrained arm
Length	long (909–2246 chars)	short (368–696)
Genre	cloud-essay vs N-ATLaS spread	all ≈ direct verdict
Commitment	much hedging/refusal	everyone commits
Language stability	DeepSeek 20/54 English, N-ATLaS 6 garbled	near-perfect
Emphatic ratio	Claude 0.324 / N-ATLaS 0.066	Claude 0.340 / N-ATLaS 0.096 / DeepSeek 0.005
What it measures	spontaneous Yoruba discourse	wrapper compliance

Methodological takeaway: the constrained arm is excellent for clean, comparable, committed answers, but its genre/commitment uniformity is **manufactured by the wrappers** — and the DeepSeek/ N-ATLaS verdict template makes their "decisiveness" non-comparable to the no-template cloud models. For claims about each model's *native* Yoruba moral voice, use the **open** arm; for claims about *content under a fixed format*, use the constrained arm. Always report which wrapper applied.

7. Bottom line (constrained)

Under the wrapper the four models converge on short, clean, committed verdicts, so the dramatic open-arm personality differences flatten. The findings that **survive** the constraint — and are thus most robust — are: **Claude's high emphatic *èmi*** (esp. trolley/whistleblower/memory) and **DeepSeek's near-total avoidance of *èmi*** (template-driven *mo*). The findings that are **artifacts of the wrapper** are the uniform direct-verdict genre and the collapse of refusals, especially for the format-forced DeepSeek and N-ATLaS. The emphatic-ratio ranking (Claude ■ GPT-4o > N-ATLaS > DeepSeek) is consistent across both arms, reinforcing that emphatic self-marking is a **stable, model-specific** trait rather than a wrapper effect.